When More Observation Hurts: Redundancy and Robust Clarification in Ask-or-Act

CS6208 Final Presentation | 5-min talk

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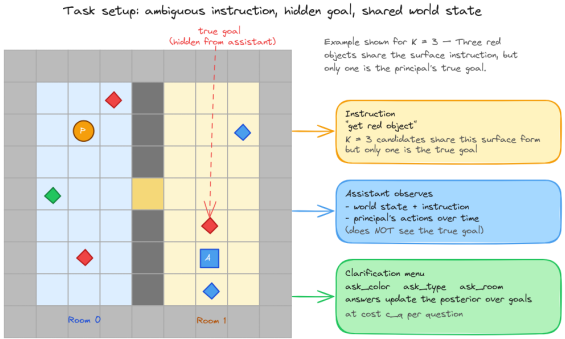
Motivation and Setup



- 9×9 two-room gridworld, K candidate goals
- Boltzmann principal (β, ε-noise)
- 6 policies, 19,800 episodes, 95% bootstrap CIs

The question

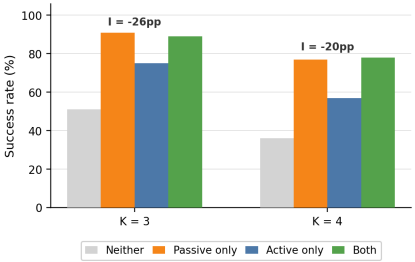
An assistant can infer the principal's goal by **passively watching** actions or **actively asking** at cost c_q.

How do these two channels interact — and when does more observation hurt?

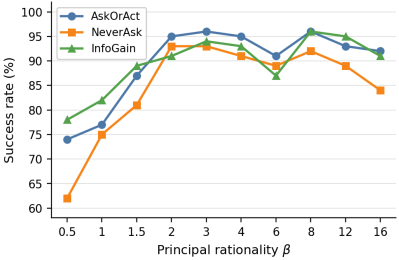
Gap

Prior work (CLIPS, ATD, ShootFirst) studies each channel in isolation. To our knowledge this is the first benchmark combining both under controlled ambiguity sweeps.

Finding 1 & 2 — Redundancy and Non-Monotonicity

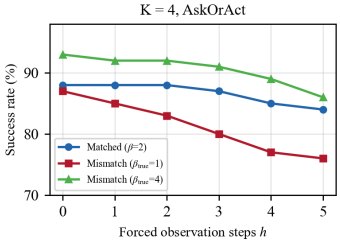


Channels are sub-additive (in this setting)
$I = -26 \pm 3$ pp at $K = 3$. Clarification matters most when passive inference is weak — +22.6 pp under full first-action aliasing.



Assistability is non-monotonic in β (in our benchmark)
NeverAsk peaks at $\beta = 8$ (92%) then drops to 84% at $\beta = 16$ as action diversity collapses. AskOrAct detects this and asks more.

Finding 3 — More Observation Can Hurt, and the Fix



Forced passive observation costs -2.2 pp/step under β-mismatch vs. -0.8 matched. Three decision-layer fixes all fail — the bottleneck is posterior quality.

AskOrAct-ST: fix the posterior

Temper passive updates by $\omega_t \in (0, 1]$ from a CUSUM surprise accumulator. Down-weight observations that are unlikely under the assumed model.

- Matched: $+0.0 \pm 1.6$ pp (no harm)
- Mismatched (β_{true}=1): $+1$–$+5$ pp

Three takeaways

1. Channels are sub-additive — exploit redundancy, don't stack.
2. Rationality can *hurt* passive inference.
3. Under mismatch, fix the **posterior**, not the gate.